

Example:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Change 0x27 to 0x3F if LCD does not work
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.init(); // Initialize LCD
  lcd.backlight(); // Turn on backlight

  lcd.setCursor(0, 0); // Column 0, Row 0
  lcd.print("Hello");
}

void loop() {
  // Nothing needed here
}
```

In-Lab Question

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define TRIG D5
#define ECHO D6
#define LED D7

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {

  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  pinMode(LED, OUTPUT);

  lcd.init();
  lcd.backlight();
  Serial.begin(9600);
}

void loop() {

  long duration;
  float distance;

  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
```

```

duration = pulseIn(ECHO, HIGH);
distance = duration * 0.034 / 2;

lcd.setCursor(0, 0);
lcd.print("Distance: ");
lcd.print(distance);

if (distance < 20) {
    digitalWrite(LED, HIGH);
    lcd.setCursor(0, 1);
    lcd.print("LED: ON ");
} else {
    digitalWrite(LED, LOW);
    lcd.setCursor(0, 1);
    lcd.print("LED: OFF ");
}

delay(500);
}

```



Post-Lab Question

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2); // Change to 0x3F if needed

char inputChar;
char outputChar;

void setup() {
    Serial.begin(9600);
    lcd.init();
    lcd.backlight();

    lcd.setCursor(0,0);
    lcd.print("Enter a char:");
}

void loop() {
    if (Serial.available() > 0) {
        inputChar = Serial.read(); // Read character
        outputChar = inputChar + 1; // Next character

        lcd.clear();

        // First row
        lcd.setCursor(0,0);
        lcd.print("INPUT: ");
        lcd.print(inputChar);

        // Second row
        lcd.setCursor(0,1);
        lcd.print("OUTPUT: ");
        lcd.print(outputChar);

        delay(500);
    }
}
```

How It Works

1. User types a character in Serial Monitor (e.g., `a`).
2. Program reads the character using `Serial.read()`.
3. ASCII value increases by 1:
 - `'a'` → `'b'`
 - `'A'` → `'B'`
 - `'1'` → `'2'`
4. LCD displays:

INPUT: a
OUTPUT: b